

$$\int \cos^m x \, dx = \frac{1}{m} \cos^{m-1} x \sin x + \frac{m-1}{m} \int \cos^{m-2} x \, dx,$$

$$\int \cos^{-m} x \, dx = \frac{1}{m-1} \cos^{-m+1} x \sin x + \frac{m-2}{m-1} \int \cos^{-m+2} x \, dx.$$